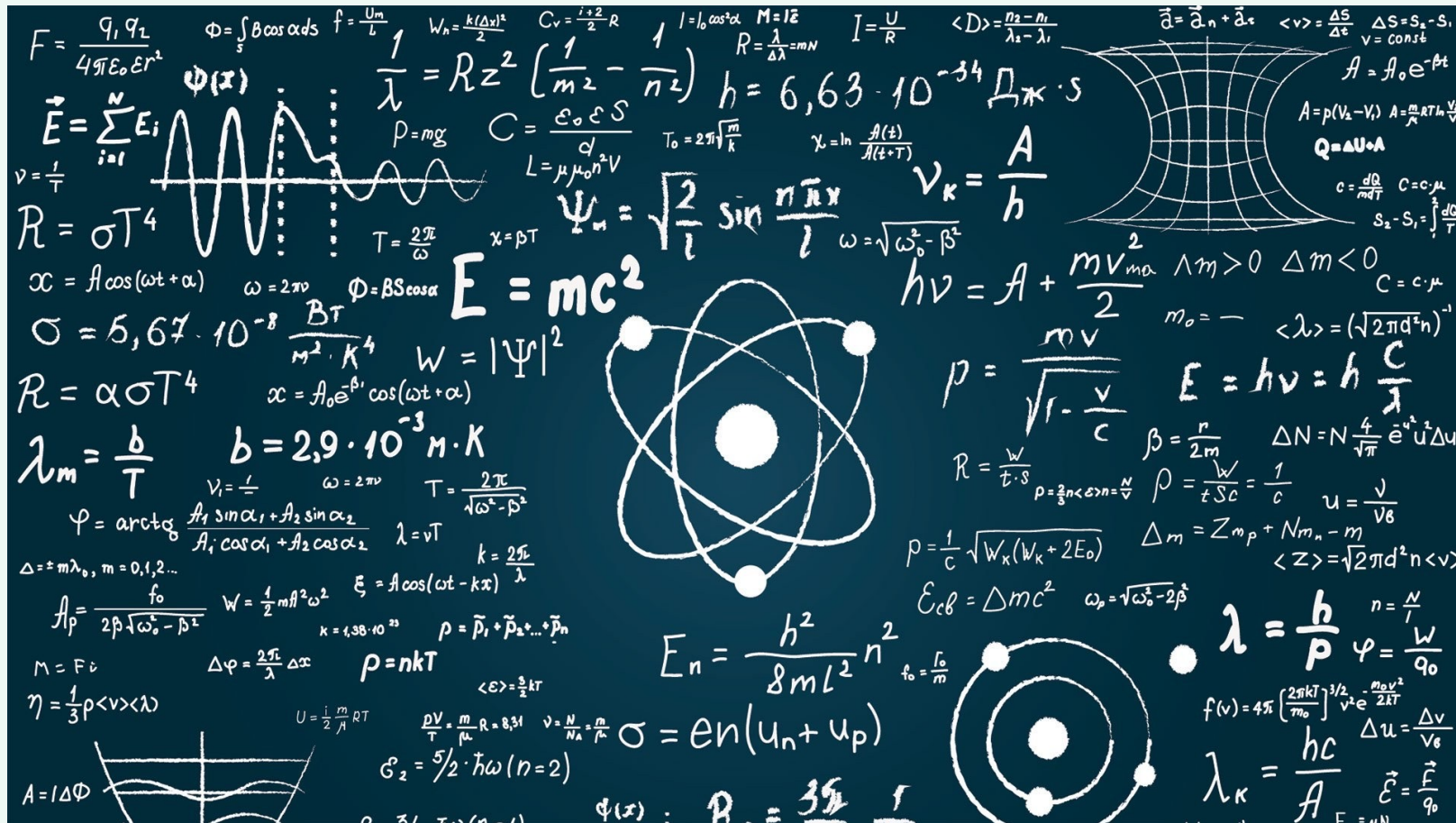


Quantum Mechanics 2



Lecture 2
(Autumn Term, 2025)

Prof Cristina Lazzeroni
c.lazzeroni@bham.ac.uk

From Lecture 1

(amplitude)² of wave function is proportional to the probability of photons being detected per unit time

Wave theory on matter particles

From Lecture 1

WAVE DESCRIPTION OF PARTICLES

[1C]

SIMPLE SINE WAVE:

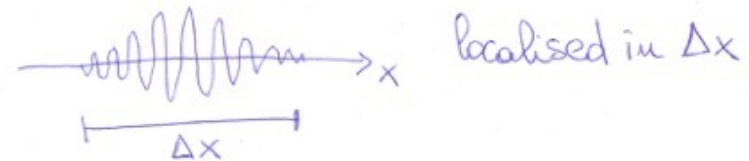


well-defined $\lambda \Rightarrow$ well defined $k = 2\pi/\lambda$

$\Rightarrow$ well-defined $p = \hbar k$

but no information about position

To describe a localised particle, need wave packet



Can create wave packet mathematically from a superposition (addition) of sine (or cosine) waves with different wavelengths and amplitudes, giving constructive interference within Δx and destructive interference outside Δx

From Lecture 1

wave packet: limited spatial range requires ^{1D} introduction of continuous range of λ

$\Rightarrow$ range of wave numbers = Δk

$\Rightarrow \Delta p$

GAIN INFORMATION ON POSITION AT
EXPENSE OF LOSING INFORMATION ON
MOMENTUM

Uncertainty on position and momentum
are related by mathematical nature of waves

$$\Delta x \Delta p \geq \frac{\hbar}{2} \quad (\text{HUP})$$

There is intrinsic limit in the product.

It doesn't depend on any particular
measurement process, it's an intrinsic limit
to quantum information.

Recap of Lecture 1

(amplitude)² of wave function is proportional to the probability of photons being detected per unit time

Wave theory on matter particles gives probabilities of particles being measured (detected) within a given range (per unit time)

Heisenberg's uncertainty principle expresses intrinsic quantum limit to the product of position and momentum uncertainties as a consequence of the wave picture

Wave picture leads to diffraction, interference etc.
BUT need equivalent of “equation of motion” for complete description of “quantum particle”.

Analogy is classical wave eqn., that is 2-order partial differential equation. In classical physics, wave eqn. can be derived from basic postulates of Newton + Maxwell.

In QM, form of wave eqn. is a basic postulate, cannot be derived from other equations.

In QM, form of wave eqn. is a basic postulate, cannot be derived from other equations.

Following a plausibility argument to justify form of Scrodinger's eqn (SE)

SE must satisfy the following constraints:

- a) must be consistent with **De Broglie's relations**
- b) must satisfy **energy conservation**
- c) **solution must be linear**, so superposition applies
- d) for a **free particle**, expect sine/cosine wavefunction

$$a) E = \hbar \omega \quad p = \hbar k \quad (1)$$

2A

$$b) \text{ Non-relativistic } E = \frac{p^2}{2m} + V(x, t) \quad (2)$$

$$\text{Combine (1) + (2): } \frac{\hbar^2 k^2}{2m} + V(x, t) = \hbar \omega \quad (3)$$

c) Wave function ψ , solution of S.E., must be linear to allow superposition:

if ψ_1 and ψ_2 are solutions, then $a\psi_1 + b\psi_2$ also is a solution (a, b constants)

$\Rightarrow$ S.E. is a linear eqn.

d) Free particle: $F = -\frac{dV}{dx} = 0 \Rightarrow V(x, t) = \text{const} = V_0$

$$\psi(x, t) = A [\cos(kx - \omega t) + \gamma \sin(kx - \omega t)]$$

$$\left. \begin{aligned} \frac{\partial \psi}{\partial t} &= A \omega [\sin(kx - \omega t) - \gamma \cos(kx - \omega t)] \\ \frac{\partial^2 \psi}{\partial x^2} &= -A k^2 [\cos(kx - \omega t) + \gamma \sin(kx - \omega t)] \end{aligned} \right\} (4) \quad \begin{array}{l} A, \gamma = \text{const} \\ \text{to be} \\ \text{found} \end{array}$$

Comparing (3) and (4), suggests a candidate QM wave equation:

$$(5) \quad \alpha \frac{\partial^2 \psi}{\partial x^2} + V_0 \psi = \beta \frac{\partial \psi}{\partial t} \quad \alpha, \beta \text{ const.}$$

Substitute (4) into (5)

$$\begin{aligned} A(-\alpha k^2 + V_0)[\cos(kx - \omega t) + \gamma \sin(kx - \omega t)] &= \\ &= A\beta\omega[\sin(kx - \omega t) - \gamma \cos(kx - \omega t)] \end{aligned} \quad (6)$$

(6) Must be valid for all (x, t) values:

$$(-\alpha k^2 + V_0 + \gamma\beta\omega)\cos(kx - \omega t) + [\gamma(V_0 - \alpha k^2) - \beta\omega]\sin(kx - \omega t) = 0$$

$\Rightarrow$ only if coefficients are equal to zero

$$-\alpha k^2 + V_0 + \gamma\beta\omega = 0 \quad (7)$$

$$\gamma(-\alpha k^2 + V_0) - \beta\omega = 0 \quad (8)$$

Substitution of (7) into (8):

$$\begin{aligned} -\alpha k^2 + V_0 &= -\gamma\beta\omega \\ \gamma(-\gamma\beta\omega) &= \beta\omega \\ -\gamma^2 &= 1 \Rightarrow \gamma = \pm i \end{aligned}$$

So $\gamma = \pm i$ sign is arbitrary, no change in physics. Choose $\gamma = +i$

Solution then is $\psi(x, t) = A[\cos(kx - \omega t) + i\sin(kx - \omega t)]$

or $\psi(x, t) = A e^{i(kx - \omega t)} \quad [\text{REMEMBER THIS}] \quad (9)$

Substitute (9) into (5) to find α, β

$$\textcircled{5} \quad \alpha \frac{\partial^2 \psi}{\partial x^2} + V_0 \psi = \beta \frac{\partial \psi}{\partial t}$$

$$\psi = A e^{i(kx - \omega t)} \rightarrow$$

$$\alpha \frac{\partial^2 \psi}{\partial x^2} + V_0 \psi = \beta \frac{\partial \psi}{\partial t} \Rightarrow \alpha i^2 k^2 \psi + V_0 \psi = -i\omega \beta \psi \quad \textcircled{10}$$

$$\text{Recall } \textcircled{3}: \quad \frac{\hbar^2 k^2}{2m} + V(x,t) = \hbar \omega$$

$\nwarrow V_0$

$$\text{Compare } \textcircled{10} \text{ with } \textcircled{3}: \quad \alpha = -\frac{\hbar^2}{2m}$$

$$\beta = i\hbar$$

Postulate: wave eqn. is valid for any $V(x,t)$

$$\boxed{\text{S.E.} \quad -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x,t)}{\partial x^2} + V(x,t) \psi(x,t) = i\hbar \frac{\partial \psi(x,t)}{\partial t}}$$

S.E. time-dependent equation

Complex 2nd-order partial-differential equation

Its solution $\psi(x,t)$ is a complex function

Summary

In QM, form of wave eqn. is a basic postulate, cannot be derived from other equations. Following a plausibility argument and satisfying constraints, we get time-dependent Schrodinger's equation:

$$\frac{-\hbar^2}{2m} \frac{\partial^2 \psi(x,t)}{\partial x^2} + V(x,t)\psi(x,t) = i\hbar \frac{\partial \psi(x,t)}{\partial t}$$

S.E. is a complex 2-order partial differential equation

Its solution is a complex function

Next lecture: implications, how to extract physical information from complex wavefunction.